ANALYTICAL CUME
APRIL 5, 2002

UV-Visible Spectroscopy in Quantitative Analysis

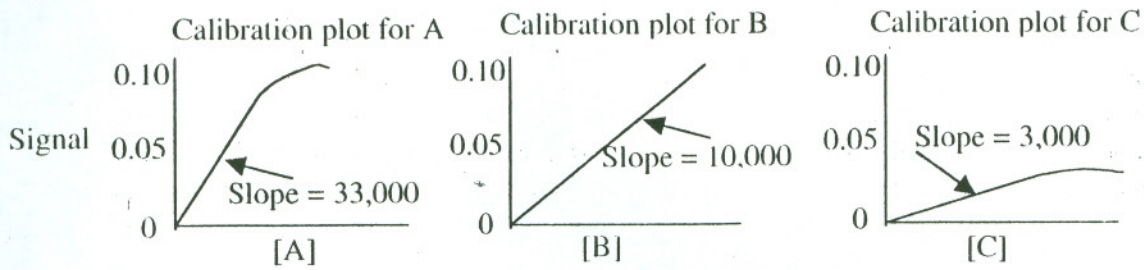
1. Standard Additions (30 points)

The chromium in an aqueous sample was determined by pipetting 10.0 mL of the unknown into each of five 50.0 mL volumetric flasks. Various volumes of a standard containing 12.2 ppm Cr were added to the flasks, following which the solutions were diluted to volume.

A least-squares fit of a plot of absorbance vs. volume of standard (in mL) gave a slope, m, of 0.00881 and a y-intercept, b, of 0.202. Calculate the ppm Cr in the sample.

2. Calibration Curves (40 points total)

An analytical method is used to obtain calibration curves for substance A (Figure A), substance B (Figure B), and substance C (Figure C). Below are some questions about these curves. Be sure to explain why you chose your answers to each of the questions.



(a) (10 points) Order the substances, A, B, and C from highest sensitivity to lowest sensitivity. Explain.

(b) (10 points) Calculate the selectivity coefficient of A relative to B.

(c) (10 points) Which substance provides the largest dynamic range? Why?

(d) (10 points) Three replicates of a blank solution were analyzed, giving the following signals: 0.1, 0.3, and 0.1. What is the detection limit for substance A? (Note: some equations are given below to help, but not all of them are relevant.)

$$\bar{x} = \frac{\sum x_i}{N}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N-1}}$$

$$S_m = \bar{S}_{blank} + k\, s_{blank}$$

$$C_m = \frac{S_m - \bar{S}_{blank}}{m}$$

$$C_x = \frac{-(V_s)_{x=0}\, C_s}{V_x}$$